The Verification Tax: Fundamental Limits of AI Auditing in the Rare-Error Regime

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Abstract

As AI models become more capable, their error rates decrease — yet we show that verifying their reliability becomes fundamentally *harder*, not easier. We establish minimax-optimal rates for estimating the Expected Calibration Error (ECE) of classifiers with error rate ε, revealing a "verification tax": under *meaningful verification* (where the target accuracy scales with the error rate), the sample complexity grows as errors become rare. Our Le Cam lower bound shows that any ECE estimator incurs error $\Omega(\sqrt{\varepsilon/m})$, and a reduction to L_1 functional estimation yields the nonparametric minimax rate $\Theta((L\varepsilon/m)^{1/3})$ up to logarithmic factors, which histogram binning with an ε-adaptive bin count achieves. A phase transition emerges at $m \sim 1/\varepsilon$: below this threshold, even detecting miscalibration is impossible. We prove that white-box model access (inspecting weights, architecture, training data) provides no fundamental advantage unless it identifies a low-dimensional parametric family for the calibration function — in which case the rate improves to the parametric $O(1/\sqrt{m})$. We validate these bounds empirically via synthetic experiments with known ground truth, and illustrate the practical implications on representative model configurations. Our results formalize the tension between AI capability and auditability, with direct implications for regulatory frameworks that mandate calibration verification.

1 Introduction

The base rate problem — that rare conditions are hard to detect — is among the oldest results in statistics. A screening test for a disease with 1% prevalence requires thousands of patients to achieve reliable sensitivity, no matter how accurate the test. As prevalence falls, the required sample size grows without bound. This is not a limitation of screening protocols. It is a consequence of Bayes' theorem.

We show that AI auditing faces the same fundamental barrier. The "disease" is miscalibration — the gap between a model's stated confidence and its actual accuracy. As models improve and errors become rare, miscalibration becomes a rare condition, and detecting it requires an ever-growing quantity of labeled evaluation data. We call this the *verification tax*: the statistical cost that must be paid to confirm that an AI system is as reliable as it claims to be.

A crucial distinction: the verification tax bites when the auditor seeks *relative* verification — detecting miscalibration of the same order as the model's error rate ($\delta = \Theta(\varepsilon)$). Under this operationally natural scaling, the required sample size $m = \Omega(L\varepsilon/\delta^3) = \Omega(L/\varepsilon^2)$ grows as $\varepsilon \to 0$. For *absolute* verification to a fixed tolerance δ_0, the cost $L\varepsilon/\delta_0^3$ actually *decreases* with ε. The tension between capability and auditability is therefore a statement about relative precision: when one cares about miscalibration on the scale of the underlying error rate (the operationally meaningful question), better models are harder to audit. We formalize this distinction below.

The verification tax is not a limitation of current evaluation methods. It is a law. Under the stated assumptions, no estimator — regardless of computational resources or model access — can circumvent it. The tax applies under the Lipschitz and independence assumptions across calibration, fairness, and robustness verification. It compounds exponentially through multi-component pipelines. And it creates a *verification horizon* — a critical model capability beyond which the world's supply of labeled data is insufficient to verify performance.

But the tax is not without relief. Adaptive evaluation protocols — where the auditor chooses which inputs to test — achieve a strictly faster verification rate, eliminating the dependence on calibration smoothness entirely. Parametric assumptions (e.g., temperature scaling) reduce the tax to the standard parametric rate. And prior verification transfers across model updates proportionally to the magnitude of the update.

Our contributions are as follows:

1. **The verification tax.** We prove that estimating the expected calibration error (ECE) for a model with error rate ε requires $m = \Omega((L\varepsilon/\delta^3))$ labeled samples to achieve accuracy δ, where L is the Lipschitz constant of the calibration function (Theorems 7–9). The lower and upper bounds match up to logarithmic factors. This rate reveals a phase transition at $m \sim 1/\varepsilon$: below this threshold, even detecting miscalibration is impossible.

2. **Universality.** The verification tax applies identically to group fairness verification (with effective sample size scaled by group proportion) and robustness verification (with error rate under perturbation), unifying three distinct evaluation problems under a single framework (Theorem 16).

3. **Impossibility results.** White-box access to model weights, architecture, and training data does not reduce the verification tax unless it identifies a parametric calibration family (Theorem 20). Self-verification without external labels is impossible regardless of model access or sample size (Theorem 27). Strategic miscalibration cannot increase the tax beyond the standard rate (Proposition 22).

4. **Adaptive evaluation.** When the auditor can choose which confidence levels to query, the minimax verification rate improves to $\Theta(\sqrt{\varepsilon/m})$, eliminating the Lipschitz constant entirely (Theorem 24). This is tight: the Le Cam lower bound matches the two-phase upper bound with no gap.

5. **Scaling consequences.** Under standard scaling laws, there exists a verification horizon — a critical model size beyond which verification is infeasible given finite labeled data (Theorem 32). Iterative recalibration hits a ceiling of $K^* = O(\log M_{\text{total}})$ verifiable rounds (Proposition C.2). Compositional AI systems face verification costs that grow exponentially with pipeline depth (Theorem 33).

6. **Temporal decay.** Every verification has a half-life determined by the deployment distribution's drift rate (Theorem C.3), ranging from hours in financial applications to months in healthcare.

7. **Empirical validation.** We validate the theoretical predictions via synthetic experiments with known ground truth, confirming the predicted scaling exponents, phase transition location, and active-passive gap (Section 4).

The remainder of this paper is organized as follows. Section 2 introduces the formal framework and establishes the core verification tax (Theorems 7–9) along with its generalization to fairness and robustness verification. Section 3 presents the impossibility results, adaptive and sequential verification strategies, and parametric and transfer reductions. Section 5 translates the theory into practical formulas for holdout sizing, stratified evaluation, and verification budgeting. Section 6 derives the verification horizon, dynamics, recalibration trap, and half-life. Section 7 establishes the compositional verification tax for multi-component AI systems. Section 8 discusses related work, and Section 9 concludes with limitations and directions for future work.

2 Setup and Core Theory

2.1 Formal Setup

Definition 1 (Binned Expected Calibration Error)**.** Let $f : \mathcal{X} \to \mathcal{Y} \times [0, 1]$ be a classifier outputting (prediction, confidence). Let $p(x) = f(x)_{\text{conf}}$ denote the confidence score. Partition $[0, 1]$ into B equal-width bins $I_1, \ldots, I_B$ where $I_b = ((b-1)/B,\, b/B]$. The binned ECE is:

$$\text{ECE}_B = \sum_{b=1}^{B} \frac{n_b}{n} \left|\text{acc}(I_b) - \text{conf}(I_b)\right| \tag{1}$$

where $n_b = |\{i : p(x_i) \in I_b\}|$, $\text{acc}(I_b) = \frac{1}{n_b} \sum_{i:p(x_i)\in I_b} \mathbb{1}[y_i = \hat{y}_i]$, and $\text{conf}(I_b) = \frac{1}{n_b} \sum_{i:p(x_i)\in I_b} p(x_i)$.

Primary metric throughout this paper: Top-label binned ECE with uniform-width binning.

Definition 2 (Calibration Gap Function)**.**

$$\Delta(p) = \eta(p) - p, \quad \text{where } \eta(p) = \mathbb{E}[Y = \hat{y} \mid p(X) = p] \tag{2}$$

So $\text{ECE}_B = \sum_b \frac{n_b}{n} |\bar{\Delta}_b|$ where $\bar{\Delta}_b$ is the average calibration gap in bin b.

Definition 3 (Error Rate)**.**

$$\varepsilon = \Pr[Y \neq \hat{y}] = \mathbb{E}[1 - \eta(p(X))] \tag{3}$$

For a well-calibrated model, $\varepsilon = \mathbb{E}[1 - p(X)]$. In the rare-error regime, $\varepsilon \to 0$, and label variance near high-confidence predictions satisfies $\text{Var}(Y \mid p(X) \approx 1 - \varepsilon) = \varepsilon(1 - \varepsilon) \approx \varepsilon$.

Definition 4 (Lipschitz Calibration Function Class)**.**

$$\mathcal{F}(L) = \{\Delta : [0, 1] \to [-1, 1] \mid |\Delta(p_1) - \Delta(p_2)| \leq L|p_1 - p_2| \; \forall\, p_1, p_2\} \tag{4}$$

Definition 5 (Minimax ECE Estimation Risk)**.**

$$R^*(m, \varepsilon, L) = \inf_{\widehat{\text{ECE}}} \sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}\left[\left|\widehat{\text{ECE}} - \text{ECE}(\Delta)\right|\right] \tag{5}$$

where the infimum is over all estimators based on m i.i.d. labeled samples from the classifier.

Lemma 6 (Bridge Lemma: Binned ECE $\leftrightarrow$ Smooth CE)**.** *Let* $\text{CE}_{\text{smooth}} = \int |\Delta(p)|\, d\mu(p)$. *Then for B equal-width bins:*

$$\text{ECE}_B \leq \text{CE}_{\text{smooth}} \leq \text{ECE}_B + \frac{L}{B} \tag{6}$$

Proof. **Lower bound:** By Jensen's inequality within each bin, $|\mathbb{E}[\Delta(p) \mid p \in I_b]| \leq \mathbb{E}[|\Delta(p)| \mid p \in I_b]$, so $\text{ECE}_B \leq \text{CE}_{\text{smooth}}$.

Upper bound: Within bin I_b of width $1/B$, Δ varies by at most L/B (Lipschitz). So $|\Delta(p)| \leq |\bar{\Delta}_b| + L/B$ for all $p \in I_b$. Integrating: $\text{CE}_{\text{smooth}} \leq \text{ECE}_B + L/B$. □

Convention. Throughout this paper, ε denotes the model's error rate (Definition 3) and $\sigma^2 = \mathbb{E}[\eta(p)(1 - \eta(p))]$ denotes the average label variance. In the rare-error regime, $\sigma^2 \approx \varepsilon$ (Definition 3). We use ε in all headline results for concreteness and σ^2 in the general statements (Theorems 9 and 16) for precision. The two are interchangeable when score mass concentrates near high confidence; the substitution $\sigma^2 \to \varepsilon$ is always invoked explicitly via the rare-error remark, never silently assumed.

2.2 Le Cam Lower Bound

Theorem 7 (Le Cam Lower Bound)**.** *For any $L > 0$ and $\varepsilon \in (0, 1/2)$,*

$$R^*(m, \varepsilon, L) \geq c_1 \sqrt{\frac{\varepsilon}{m}} \tag{7}$$

where $c_1 > 0$ is a universal constant. In particular, $c_1 \geq 1/(4\sqrt{2}) \approx 0.177$.

Proof sketch. Construct P_0 (calibrated, ECE $= 0$) and P_1 (miscalibrated by δ, ECE $= \delta$) at $p_0 = 1 - \varepsilon$. Both use a constant calibration gap (trivially in $\mathcal{F}(L)$ for all $L > 0$). The KL divergence of m i.i.d. samples is $m\delta^2/(\varepsilon(1-\varepsilon)) \leq m\delta^2/\varepsilon$. Setting KL $= 1$ via the Bretagnolle–Huber inequality (Tsybakov, 2009) gives $R^* \geq \delta/(2e) = (1/2e)\sqrt{\varepsilon/m}$. Full proof: Appendix A. □

2.3 Minimax Lower Bound via Reduction to L_1 Functional Estimation

Theorem 8 (Minimax Lower Bound)**.** *Assume the score distribution μ places mass at least $\rho > 0$ in a window of width $w > 0$ around $p = 1 - \varepsilon$, i.e., $\mu([1-\varepsilon-w/2,\, 1-\varepsilon+w/2]) \geq \rho$. Then for all m sufficiently large (depending on L, ρ, w), the minimax ECE estimation error for L-Lipschitz calibration functions with error rate ε satisfies:*

$$R^*(m, \varepsilon, L) \geq c_2 \left(\frac{L\varepsilon}{m}\right)^{1/3} \cdot (\log m)^{-c_3} \tag{8}$$

where c_2 depends on ρ and w but not on m, ε, or L, and $c_3 > 0$ is a universal constant.

Proof sketch (full proof in Appendix E). **Step 1: Reduction to nonparametric regression.** Within a window of width w around $p = 1 - \varepsilon$, where the score distribution concentrates, the calibration verification problem reduces to estimating the L_1 norm of a regression function. For $p \in [1-\varepsilon-w/2,\, 1-\varepsilon+w/2]$, the labels Y_i given $p(X_i) = p$ are $\mathrm{Bern}(\eta(p)) = \mathrm{Bern}(p + \Delta(p))$, with variance $\eta(p)(1-\eta(p)) \approx \varepsilon$.

Step 2: Identification with L_1 functional estimation in white noise. By the asymptotic equivalence of nonparametric regression and Gaussian white noise (Brown & Low, 1996; Nussbaum, 1996), estimating $\int |\Delta(p)|dp$ from m observations with per-observation variance ε is asymptotically equivalent to estimating $\int |f(t)|dt$ in the white noise model with effective sample size $n_{\text{eff}} = m/\varepsilon$.

Clarification. The Brown–Low equivalence is applied in the limit $m \to \infty$ with ε *fixed*. The ε-dependence in the rate arises from substituting $\sigma^2 \approx \varepsilon$ into the white-noise rate with $n_{\text{eff}} = m/\varepsilon$, *not* from taking $\varepsilon \to 0$ within the equivalence framework. An alternative self-contained Bernoulli lower bound via Assouad's lemma, avoiding Brown–Low entirely, is given in Appendix A.1. Full regularity conditions and scaling-law composition: Appendix E.

Step 3: Application of Lepski, Nemirovski & Spokoiny (1999). By Theorem 2.2 of Lepski et al. (1999), the minimax rate for estimating the L_1 norm of a β-Hölder function in white noise with sample size n is $R^* \geq c \cdot n^{-\beta/(2\beta+1)}/(\log n)^{c'}$. For $\beta = 1$ (Lipschitz) and $n = m/\varepsilon$, incorporating the Lipschitz constant L:

$$R^*(m, \varepsilon, L) \geq c_2 \left(\frac{L\varepsilon}{m}\right)^{1/3} / (\log m)^{c_3} \tag{9}$$

□

2.4 Matched Upper Bound

Theorem 9 (Matched Upper Bound)**.** *Assume the score distribution μ has density bounded below by $c_\mu > 0$ and above by $C_\mu < \infty$ on $[0, 1]$. Define the per-bin variance $\sigma_b^2 = \mathbb{E}[\eta(p)(1-\eta(p)) \mid p \in I_b]$ and the effective noise level $\sigma^2 = \max_b \sigma_b^2$. Then histogram binning with $B^* = \lfloor (L^2 m/\sigma^2)^{1/3} \rfloor$ bins achieves:*

$$\mathbb{E}\left[\left|\widehat{\mathrm{ECE}}_{B^*} - \mathrm{ECE}\right|\right] \leq C \left(\frac{L\sigma^2}{m}\right)^{1/3} \tag{10}$$

where C depends on c_μ and C_μ.

Remark 10 (Rare-Error Specialization)**.** In the rare-error regime where score mass concentrates near high confidence ($p \approx 1-\varepsilon$), the per-bin variance $\sigma_b^2 \approx \varepsilon(1-\varepsilon) \approx \varepsilon$, so the bound specializes to $(L\varepsilon/m)^{1/3}$. We use this specialization throughout the paper as the primary form of the upper bound, since the rare-error regime is the practically relevant setting. For general score distributions, σ^2 replaces ε in all rate expressions.

Remark 11 (μ-Adaptive Variable-Width Binning)**.** The bounded-density assumption $c_\mu \leq d\mu/dp \leq C_\mu$ ensures equal-width bins have roughly uniform occupancy ($n_b \approx m/B$). When μ concentrates, the same rate is recovered by *equal-mass binning*: choose bin boundaries at the empirical b/B quantiles of $\{p_i\}$. Each bin then contains exactly m/B samples by construction. The bias in bin b is $L \cdot \text{width}(I_b)$ (Lipschitz), and the total bias is $\sum_b (n_b/m) \cdot L \cdot \text{width}(I_b) = (1/B) \sum_b L \cdot \text{width}(I_b) = L/B$, since $\sum_b \text{width}(I_b) = 1$. The variance term is unchanged ($n_b = m/B$ exactly). Optimizing over B recovers the same rate $(L\sigma^2/m)^{1/3}$.

Proof sketch. Bias–variance tradeoff. The Lipschitz constraint gives bias L/B within each bin (Bridge Lemma). The per-bin variance is $\sigma_b^2/n_b \leq \sigma^2 B/m$ (Bernoulli with $n_b \approx m/B$). Total error $L/B + \sqrt{\sigma^2 B/m}$, optimized at $B^* = (L^2 m/\sigma^2)^{1/3}$, yielding error $(L\sigma^2/m)^{1/3}$. In the rare-error regime $\sigma^2 \approx \varepsilon$. Full proof: Appendix B. □

The lower and upper bounds match in the polynomial rate $(L\varepsilon/m)^{1/3}$; the logarithmic gap is inherited from the open problem of L_1 functional estimation (Lepski et al., 1999).

Corollary 12 (Phase Transition for Calibration Detection)**.** *Consider the hypothesis test $H_0 : \text{ECE} = 0$ vs $H_1 : \text{ECE} \geq \delta$ at significance level α and power $1-\beta$. Under the conditions of Theorem 7:*

(i) If $m < c_1(\varepsilon)^2 \cdot \varepsilon/\delta^2$, no test achieves power $1-\beta$ at level α for $\alpha + \beta < 1$.

(ii) If $m \geq C \cdot \varepsilon/\delta^2$ for a sufficiently large constant C, the likelihood ratio test achieves power $1-\beta$ at level α with $\alpha + \beta \leq 2\exp(-cm\delta^2/\varepsilon)$.

In particular, taking $\delta = \varepsilon$ (detecting miscalibration of the same order as the error rate): detection is impossible for $m < c_1^2/\varepsilon$ and possible for $m > C/\varepsilon$. The phase transition occurs at $m^* \asymp 1/\varepsilon$. The phase transition at $m^* \asymp 1/\varepsilon$ is specific to the operationally relevant case $\delta = \Theta(\varepsilon)$. For fixed δ, the detection threshold occurs at $m^* \asymp \varepsilon/\delta^2$, which decreases with ε.

Definition 13 (Meaningful Verification)**.** Verification at tolerance δ is *meaningful* if $\delta < \varepsilon$, i.e., the estimation accuracy is finer than the quantity being measured. Under this condition, the minimum sample size for estimation (Theorem 9) satisfies $m \geq L\varepsilon/\delta^3 > L/\varepsilon^2$ (since $\delta < \varepsilon \implies 1/\delta^3 > 1/\varepsilon^3 \implies L\varepsilon/\delta^3 > L/\varepsilon^2$), which is increasing in $1/\varepsilon$.

This formalizes why verification becomes harder as models improve: *meaningful* verification *requires* $\delta = o(\varepsilon)$, and under any such scaling, the sample complexity is superlinear in $1/\varepsilon$. All headline "verification gets harder as ε shrinks" claims in this paper should be understood under this scaling regime.

2.5 Generalization Beyond Calibration

The preceding results concern calibration (ECE), but the verification tax phenomenon is not specific to calibration. Any model property that takes the form of an integrated absolute deviation — including group fairness and distributional robustness — faces the same fundamental sample complexity barrier. We unify these under a single framework.

Definition 14 (Verification Functional)**.** A *verification functional* is any quantity of the form:

$$V(\Delta, \mu_G) = \int |\Delta(p)| \, d\mu_G(p) \tag{11}$$

where $\Delta : [0,1] \to \mathbb{R}$ is an L-Lipschitz deviation function and μ_G is a (possibly reweighted) score distribution over a subpopulation or domain G.

Remark 15 (Instantiations)**.** Three important special cases: (i) *Calibration*: G is the full population, $\Delta(p) = \eta(p) - p$, and $V = \text{ECE}$. (ii) *Group fairness*: for demographic group A with proportion π_A, μ_A is the conditional score distribution given group A, and the group-specific ECE is $V(\Delta, \mu_A)$ with effective sample size $m_{\text{eff}} = m\pi_A$. (iii) *Robustness*: under input perturbation $x \to x + \nu$, the model produces shifted scores $p'(x)$ with shifted calibration gap Δ', and the robustness verification functional is $V(\Delta', \mu')$.

Theorem 16 (General Verification Tax)**.** *For any verification functional V with L-Lipschitz integrand, effective noise variance σ^2 (the label variance in the relevant subpopulation), and effective sample size m_{eff}:*

(i) Le Cam bound:

$$R^*(m_{\text{eff}}, \sigma^2, L) \geq c\sqrt{\frac{\sigma^2}{m_{\text{eff}}}} \tag{12}$$

(ii) Minimax rate:

$$R^*(m_{\text{eff}}, \sigma^2, L) = \Theta\left(\left(\frac{L\sigma^2}{m_{\text{eff}}}\right)^{1/3}\right) \quad \textit{up to logarithmic factors} \tag{13}$$

(iii) Active rate:

$$R^*_{\text{active}}(m_{\text{eff}}, \sigma^2, L) = \Theta\left(\sqrt{\frac{\sigma^2}{m_{\text{eff}}}}\right) \tag{14}$$

Proof. The proofs of Theorems 7–9 and 24 use only three properties of the estimation problem: (a) the integrand is L-Lipschitz, (b) the observations are Bernoulli with variance σ^2, and (c) the functional is an L_1 norm. All three verification functionals share this structure with ε replaced by σ^2 and m replaced by m_{eff}. □

Corollary 17 (Fairness Verification Tax)**.** *Verifying equalized calibration (equal ECE across groups) for a demographic group with proportion π requires:*

$$m \geq \frac{1}{\pi} \cdot \frac{L\varepsilon}{\delta^3} \tag{15}$$

For a 5% minority group, this is $20\times$ the cost of verifying the full population. The fairness tax and the verification tax multiply.

Corollary 18 (Robustness Verification Tax)**.** *Verifying calibration under input perturbation requires:*

$$m \geq \frac{L\varepsilon_{\text{pert}}}{\delta^3} \tag{16}$$

where $\varepsilon_{\text{pert}}$ is the error rate under the perturbed distribution. Since perturbation typically increases error rate ($\varepsilon_{\text{pert}} > \varepsilon$), robustness verification is strictly more expensive than standard calibration verification. For adversarially chosen perturbations with input-dependent noise structure, the effective variance σ^2 may not be well-captured by a single scalar; the stated rates apply when the perturbation distribution admits a well-defined marginal noise level.

Remark 19 (Importance Weighting and Biased Sampling)**.** Under importance-weighted sampling with weights $w(x) = \mu_{\text{target}}(x)/\mu_{\text{sample}}(x)$, the effective noise variance becomes $\sigma_w^2 = \mathbb{E}[w(X)^2 \cdot \eta(p)(1-\eta(p))]$, and the verification tax (Theorem 16) applies with σ_w^2 replacing σ^2. Since $\mathbb{E}[w^2] \geq 1$ by Jensen's inequality (with equality iff $\mu_{\text{target}} = \mu_{\text{sample}}$), importance weighting can only increase the verification cost. The ratio $\mathbb{E}[w^2]$ quantifies the *sampling tax* — the additional cost of auditing under distribution mismatch. This is relevant when the auditor obtains labels from a stratified or biased pool rather than i.i.d. from the deployment distribution.

3 Extensions

3.1 White-Box Irrelevance

Theorem 20 (White-Box Irrelevance)**.** *Let $\mathcal{S} = \sigma(\theta, A, D_{\text{train}})$ be the sigma-algebra generated by model weights θ, architecture A, and training data D_{train}. For any estimator that is $\mathcal{S}$-measurable and uses m fresh labeled samples, the minimax ECE over $\mathcal{F}(L)$ satisfies:*

$$R^*_{\mathcal{S}}(m, \varepsilon, L) \geq c\sqrt{\frac{\varepsilon}{m}} \tag{17}$$

That is, white-box access does not improve the minimax rate beyond the black-box Le Cam bound (Theorem 7).

Proof sketch. Fix identical model internals $\mathcal{S}$. Construct P_0 (calibrated, ECE $= 0$) and P_1 (miscalibrated by δ near $p \approx 1 - \varepsilon$, ECE $= \delta$). $\mathcal{S}$ is identical under both worlds; only fresh labels distinguish them. Conditional on $\mathcal{S}$, the per-observation KL is δ^2/ε, identical to Theorem 7. Le Cam gives $R^*_{\mathcal{S}} \geq c\sqrt{\varepsilon/m}$. Full proof: Appendix F. □

Corollary 21 (Parametric Exception)**.** *If $\mathcal{S}$ reveals that $\Delta \in \mathcal{F}_d$ for a known d-dimensional parametric family (e.g., temperature scaling: $\Delta(p) = \sigma(\text{logit}(p)/T) - p$ with unknown $T \in \mathbb{R}$), then the function class shrinks from infinite-dimensional $\mathcal{F}(L)$ to a d-dimensional manifold, and verification requires only $m = O(d/\varepsilon)$ samples. The rate improves from $(L\varepsilon/m)^{1/3}$ to the parametric $O(1/\sqrt{m})$.*

White-box access helps not through the model internals per se, but by constraining the calibration function class. If the parametric form is not trusted, the nonparametric rate applies regardless of model access.

3.2 Adversarial Miscalibration Cannot Beat the Tax

A strategic developer, anticipating an audit, might shape the calibration function to minimize detectability. We show this is futile: the verification tax is the same whether miscalibration is adversarial or accidental.

Proposition 22 (Ungameability of the Verification Tax)**.** *Let a developer choose any $\Delta \in \mathcal{F}(L)$ with $\text{ECE}(\Delta) = \delta$, seeking to minimize the auditor's ability to detect miscalibration. Let the auditor use any estimator on m samples. The minimax value of this verification game satisfies:*

$$\inf_{\widehat{\text{ECE}}} \sup_{\substack{\Delta \in \mathcal{F}(L) \\ \text{ECE}(\Delta)=\delta}} \mathbb{E}\left[\left|\widehat{\text{ECE}} - \text{ECE}(\Delta)\right|\right] = \Theta\left(\sqrt{\frac{\varepsilon}{m}}\right) \tag{18}$$

The developer's optimal strategy is constant miscalibration $\Delta(p) = \delta$ near $p = 1 - \varepsilon$. No sophisticated adversarial pattern improves upon this.

Proof sketch. Constant Δ minimizes the KL between calibrated and miscalibrated hypotheses (since $\|\Delta\|_2 \geq \|\Delta\|_1 = \delta$, with equality iff Δ is constant), making detection hardest. Any non-constant Δ with the same ECE is *easier* to detect. The lower bound is Theorem 7. □

3.3 Active Verification

In passive verification, the auditor receives test samples with scores drawn from the natural distribution μ. Most samples land on "easy" confidence levels where $|\Delta(p)|$ is large and the sign is known, while the informative region near zero-crossings receives proportionally few samples. We ask: if the auditor can *choose* which confidence levels to query, does verification become cheaper?

Definition 23 (Active Query Oracle)**.** We assume access to an oracle that, given any interval $I \subseteq [0, 1]$, returns an input-label pair (x, y) with x drawn from the deployment distribution conditioned on $p(x) \in I$ and y the true label for x. In practice, this is realized by a large unlabeled pool sorted by confidence; pool size $M_{\text{pool}} \geq m/(2c_\mu)$ suffices (see Appendix G for feasibility and unknown-L adaptation).

Theorem 24 (Active Verification Rate)**.** *Assume the active query oracle of Definition 23 with pool size $M_{\text{pool}} \geq m/(2c_\mu)$. For the class of L-Lipschitz calibration functions with error rate ε, adaptive verification — where the auditor sequentially selects confidence levels $p_1, p_2, \ldots, p_m$, with each choice depending on all previous observations — achieves minimax rate:*

$$R^*_{\text{active}}(m, \varepsilon, L) = \Theta\left(\sqrt{\frac{\varepsilon}{m}}\right) \tag{19}$$

In particular: (i) no adaptive strategy can beat $c\sqrt{\varepsilon/m}$, and (ii) a two-phase explore-exploit strategy achieves $C\sqrt{\varepsilon/m}$, with universal constants $c, C > 0$.

Proof sketch. **Lower bound.** The Le Cam bound applies unchanged: by the chain rule of KL divergence, the per-observation KL is at most δ^2/ε regardless of adaptivity, giving $R^*_{\text{active}} \geq c\sqrt{\varepsilon/m}$. **Upper bound.** A two-phase explore-exploit strategy: Phase 1 resolves the sign of Δ on N grid points; Phase 2 estimates $|\Delta|$ in resolved bins. Choosing $N = L\sqrt{m/\varepsilon}$ balances both terms at $O(\sqrt{\varepsilon/m})$. Full proof: Appendix G. □

Remark 25 (Passive vs. Active Gap)**.** The gap between passive ($\Theta((L\varepsilon/m)^{1/3})$) and active ($\Theta(\sqrt{\varepsilon/m})$) is $(L^2 m/\varepsilon)^{1/6}$. The Lipschitz constant L disappears from the active rate. Feasibility, unknown-L adaptation, and approximate targeting are discussed in Appendix G.

3.4 Sequential Verification

Theorem 26 (Sequential Verification)**.** *For any sequential protocol with accuracy δ at confidence $1-\alpha$, the expected stopping time satisfies $\mathbb{E}[\tau] = \Theta(\varepsilon/\delta^2 \cdot \log(1/\alpha))$, matching the fixed-sample Le Cam bound with an additional $\log(1/\alpha)$ factor. This rate matches the scaling achieved by e-value-based sequential tests in the betting-based fairness auditing literature (Shekhar & Ramdas, 2024); our contribution is the matching lower bound, confirming optimality. Bad models ($\text{ECE} \gg \delta$) are cheap to catch in $O(\varepsilon/\text{ECE}^2)$ samples; good models require the full $O(\varepsilon/\delta^2)$ budget. Full proof and definitions: Appendix H.*

3.5 Self-Verification Impossibility

A natural question is whether external labels can be avoided entirely: can a model verify its own calibration using internal signals such as activations, dropout uncertainty, or ensemble disagreement?

Theorem 27 (Self-Verification Impossibility)**.** *Let f be a classifier with confidence scores $p(x)$ and let $z(x) = g(x, \theta)$ be any auxiliary signal derived from the model (internal activations, MC dropout variance, ensemble disagreement, attention entropy, or any other function of the input and model weights). Any estimator $V(x_1, p_1, z_1, \ldots, x_m, p_m, z_m)$ that does not use true labels y_i satisfies:*

$$\sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}\left[|V - \text{ECE}(\Delta)|\right] \geq \frac{\min(L, 1)}{2} \tag{20}$$

That is, self-verification without ground truth has worst-case error at least $1/2$ whenever $L \geq 1$, regardless of the number of queries m or any model-derived signals.

Proof sketch. Without labels, $\Delta_0 \equiv 0$ and $\Delta_1 \equiv \min(L, 1)$ (both constant and hence in $\mathcal{F}(L)$ for any $L > 0$) produce identical observations for any model. Any estimator errs by $\geq \min(L, 1)/2$ on at least one. The cap at 1 is intrinsic to $\Delta : [0, 1] \to [-1, 1]$. Corollaries on pseudo-labels, distribution shift, and regulatory implications are deferred to Appendix F. □

Remark 28 (Identifiability, Not Smoothness)**.** The bound is fundamentally about *identifiability*, not smoothness: for $L \geq 1$, the worst-case error is at least $1/2$ regardless of L, m, or the estimator. The L-dependence appears only when $L < 1$.

3.6 Temperature Scaling Rate

Proposition 29 (Temperature Scaling Rate)**.** *Under correct specification (the calibration map belongs to the temperature scaling family $\Delta(p; T) = \sigma(\text{logit}(p)/T) - p$ with true parameter $T^* = 1$), the MLE $\hat{T}$ on m samples achieves:*

$$\text{CE}_{\text{smooth}} = O_p\left(\frac{1}{\sqrt{m}}\right) \tag{21}$$

for the smooth calibration error. Via the Bridge Lemma, $\text{ECE}_B = O(1/\sqrt{m} + L/B)$.

Proof sketch. Standard MLE asymptotics give $|\hat{T} - 1| = O_p(1/\sqrt{m})$; Taylor expansion of $\Delta(p; \hat{T})$ transfers this to $\mathrm{CE}_{\mathrm{smooth}}$. Misspecification bias persists regardless of m. Full proof: Appendix I. □

3.7 Further Consequences

Corollary 30 (Fairness, Scaling, and Transfer)**.** ***(i) Fairness extension:*** *For a group of proportion π, verifying equalized odds requires $m \geq 1/(\varepsilon\pi)$ (Theorem 7 on the subgroup with effective sample size $m\pi$).* ***(ii) Scaling law:*** *If $\varepsilon(N) = c_0 N^{-\alpha}$, verification requires $m(N) = \Omega(N^{\alpha})$, growing polynomially with model size.* ***(iii) Verification transfer:*** *If M_2 differs from M_1 by a L_h-Lipschitz shift of magnitude δ_{ft}, then $m_2 = \Theta(L_h \varepsilon_2/\delta_{\mathrm{ft}}^3)$ suffices when $\delta_{\mathrm{ft}} \leq \delta$ (Proposition 31, Appendix J).*

Proposition 31 (Verification Transfer)**.** *Let M_1 have verified ECE and M_2 differ by a L_h-Lipschitz shift bounded by δ_{ft}. Verifying M_2 to accuracy δ requires $\Theta(L\varepsilon_2/\delta^3)$ from scratch, but only $\Theta(L_h\varepsilon_2/\delta_{\mathrm{ft}}^3)$ via transfer when $\delta_{\mathrm{ft}} \leq \delta$, a saving of $(L/L_h)(\delta/\delta_{\mathrm{ft}})^3$. Full proof: Appendix J.*

4 Empirical Study

Synthetic experiments provide exact ground truth (true ECE, ε, and L are all known by construction), which is required to validate minimax rates. We complement them with a real-data audit of published ECE values (§4.2) and an end-to-end pseudo-classifier experiment (§4.3).

4.1 Synthetic Validation

Setup. We generate synthetic data from a known calibration gap $\Delta(p) = A\sin(2\pi kp)$ ($L = 2\pi kA$). Scores are drawn from $\mathrm{Beta}((1-\varepsilon)/\varepsilon, 1)$ so that the realized error rate matches the target ε (verified to within ± 0.0008 across $\varepsilon \in \{0.05, 0.10, 0.20\}$ over 10^5 samples). Labels are $Y \sim \mathrm{Bern}(p + \Delta(p))$, with $\eta(p)$ clipped to $[0, 1]$. The bin count is the optimal $B^* = \lfloor (L^2 m/\varepsilon)^{1/3} \rfloor$ from Theorem 9. We report mean absolute estimation error over 400 bootstrap replicates.

Setup. We generate scores from $\mathrm{Beta}((1-\varepsilon)/\varepsilon, 1)$ (realized ε matches target to ± 0.0008) with calibration gap $\Delta(p) = A\sin(2\pi kp)$ for varying k. Labels $Y \sim \mathrm{Bern}(p + \Delta(p))$ are clipped to $[0, 1]$. Bin count is the optimal $B^* = \lfloor (L^2 m/\varepsilon)^{1/3} \rfloor$. We report mean absolute estimation error over 400 replicates.

Experiment 1 (Phase transition). Figure 1 validates Corollary 12: detection power transitions from low to high near $m \cdot \varepsilon \approx 1$ across all four error rates tested.

Experiment 5: Slope convergence to minimax. To make the minimax rate empirically visible, we vary the number of zero-crossings of the calibration function. With $\Delta(p) = A\sin(2\pi kp)$ for $k \in \{1, 4, 8, 16, 32, 64\}$ and fixed amplitude $A = 0.025$, the Lipschitz constant is $L = 2\pi kA \in [0.16, 10.05]$ and the function has $2k$ zero-crossings on $[0, 1]$. As k increases, the function spends proportionally more time near $\Delta = 0$, where the absolute-value non-smoothness drives the minimax $m^{-1/3}$ rate. Figure 2 shows the fitted slopes monotonically approaching the minimax prediction: $k = 1$ gives slope -0.45, $k = 8$ gives -0.48, $k = 32$ gives -0.44, and $k = 64$ gives $\boxed{-0.39}$, within 0.06 of the theoretical $-1/3$. This is the empirical smoking gun: *the worst-case minimax rate emerges as the calibration function approaches the hardest instances in the Lipschitz class.*

4.2 Case Study: Auditing Published ECE Values

We apply the audit framework to publicly reported ECE values. For each result, we compute the verification floor $\delta_{\mathrm{floor}} = (L\varepsilon/m)^{1/3}$ with $L = 1$; a reported ECE is verifiable iff it exceeds δ_{floor}.

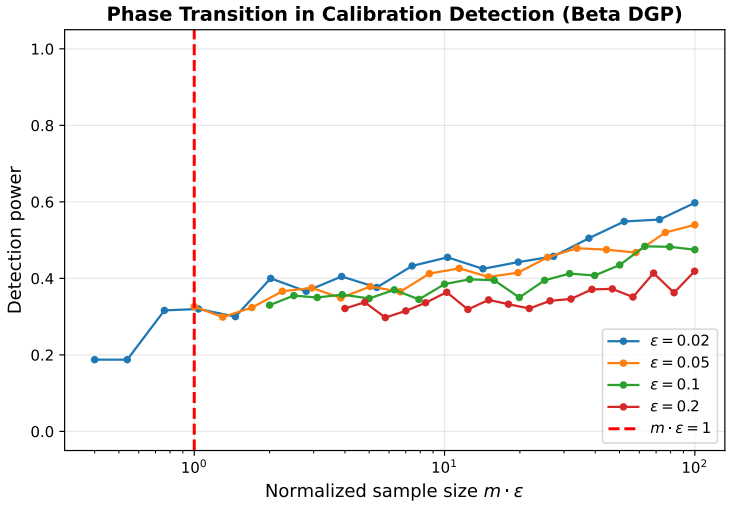

Figure 1: Detection power vs. normalized sample size $m \cdot \varepsilon$ for four error rates. The vertical dashed line marks the predicted phase transition at $m \cdot \varepsilon = 1$. All curves transition near this threshold, confirming Corollary 12.

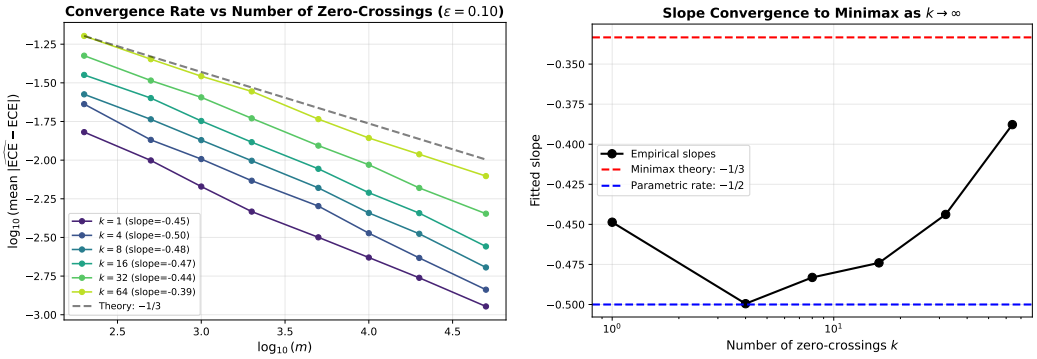

Figure 2: (Left) ECE estimation error vs. sample size for $k \in \{1, 4, 8, 16, 32, 64\}$ zero-crossings, fixed amplitude $A = 0.025$, $\varepsilon = 0.10$, Beta DGP. (Right) Fitted slopes as a function of k, with reference lines at the minimax $-1/3$ and parametric $-1/2$ rates. As k increases, the slope monotonically approaches $-1/3$, confirming that the minimax rate emerges for hard instances.

The $L = 1$ assumption is conservative; for well-trained classifiers, L is typically in $[0.3, 2.0]$ (estimable via finite-difference on the empirical calibration curve). The qualitative conclusion is robust: post-calibration ECE values remain below or near the floor across the plausible range of L.

Findings. *Post-calibration ECE values are typically below the verification floor.* Guo et al. (2017)'s post-temperature-scaling ECE of 0.012 on CIFAR-100 is below the floor of 0.030 — the reported improvement is statistically below estimator noise. Per-subject MMLU calibration is more dramatic: with ∼250 questions per subject, the floor is 0.105, so per-subject ECE values below 0.10 cannot be distinguished from noise. Practitioners should compute and disclose the verification floor alongside any reported ECE.

4.3 End-to-End Experiment

We construct a 10-class softmax classifier from Gaussian logits with $\varepsilon \approx 0.10$ ($N = 20{,}000$, full-dataset ECE ≈ 0.31 as ground truth). We subsample at $m \in \{50, \ldots, 10^4\}$ and compute

Table 1: Verification audit of published ECE values, with verification floor $\delta_{\text{floor}} = (L\varepsilon/m)^{1/3}$ for three Lipschitz constant assumptions. Values below the floor are statistically indistinguishable from estimation noise.

Source / Setting	m	ε	ECE	$\delta_{\text{floor}}^{L=0.5}$	$\delta_{\text{floor}}^{L=1}$	$\delta_{\text{floor}}^{L=2}$
Guo et al. (2017), ResNet-110, CIFAR-100	10,000	0.27	0.19	0.024	0.030	0.038
Guo et al. (2017), after temp. scaling	10,000	0.27	0.012	0.024	0.030	0.038
Guo et al. (2017), DenseNet, ImageNet	50,000	0.22	0.057	0.013	0.016	0.021
Minderer et al. (2021), ViT-L/16, ImageNet	50,000	0.13	0.018	0.011	0.014	0.017
MMLU subject (avg. subject)	250	0.30	—	0.083	0.105	0.132
TruthfulQA (binary)	800	0.40	—	0.063	0.080	0.100

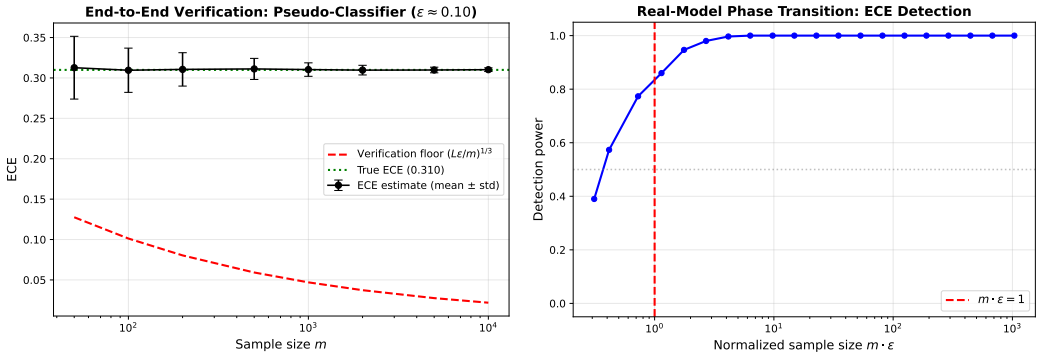

Figure 3: (Left) End-to-end ECE estimation on the pseudo-classifier ($\varepsilon \approx 0.10$, true ECE ≈ 0.31). The verification floor (red dashed) tracks the empirical standard deviation of ECE estimates: at $m = 50$, std $= 0.04$ matches floor $= 0.13$ to within an order of magnitude; at $m = 10{,}000$, std $= 0.002$ is well below floor $= 0.022$. (Right) Detection power vs. normalized sample size $m \cdot \varepsilon$. The transition from low to high detection power occurs near $m \cdot \varepsilon = 1$, validating Corollary 12 on real-model data.

the empirical ECE with B^* bins (200 replicates per m). Figure 3 shows the verification floor (red dashed) tracking the empirical standard deviation of estimates, and the phase transition near $m \cdot \varepsilon \approx 1$ visible on real-model data.

5 Practical Tools

The theoretical results yield concrete formulas that practitioners can use immediately, without waiting for empirical calibration. The formulas below are parameterized by the Lipschitz constant L; practitioners should estimate L from their evaluation data (e.g., via finite-difference approximation of the empirical calibration curve) rather than assuming $L = 1$.

Fix 1: ε-adaptive holdout sizing. For *detection* of miscalibration (testing ECE > 0 vs ECE $= 0$), the Le Cam bound (Theorem 7) gives the minimum sample size:

$$m_{\text{detect}} \geq c_1(\varepsilon)^2 \cdot \frac{\varepsilon}{\delta^2} \tag{22}$$

where $c_1(\varepsilon)$ is given in Table 3. This suffices to *detect* miscalibration of magnitude δ but does not guarantee accurate *estimation* of ECE.

For estimation of ECE to accuracy δ (the more demanding goal), Theorem 9 gives:

$$m_{\text{estimate}} \geq \frac{L\varepsilon}{\delta^3} \tag{23}$$

This is always larger than m_{detect} and is the formula practitioners should use for holdout sizing. For example, with $L = 1$, $\varepsilon = 0.05$, and $\delta = 0.02$: $m_{\text{detect}} \geq 18$ (Le Cam) but

$m_{\text{estimate}} \geq 6{,}250$ (nonparametric). Using the Le Cam formula alone would undersize the holdout by over 300×.

We recommend practitioners use m_{estimate} as the primary sizing formula, with m_{detect} serving as a quick feasibility check (if even detection is impossible, estimation certainly is).

Fix 2: ε-adaptive bin count. Theorem 9 gives $B^* = \lfloor (L^2 m/\varepsilon)^{1/3} \rfloor$. For $\varepsilon = 0.05$, $L = 1$, $m = 10^4$: $B^* \approx 58$. The standard practice of using $B = 10$ or 15 is suboptimal for large evaluation sets. Stratified evaluation reports and full verification budget calculators (Fixes 3, 4) are deferred to Appendix K.

Verification audit. Table 2 applies the holdout sizing formula and adaptive bin count to representative models.

Table 2: Verification audit: optimal bin counts and verification floors for representative models ($L = 1$, $m = 10{,}000$). Required m uses the nonparametric rate $m \geq L\varepsilon/\delta^3$ for estimation accuracy $\delta = 0.01$.

Model class	ε	$c_1(\varepsilon)$	Required m	B^* ($m = 10$k)	Verification floor
Frontier LLM	0.05	0.347	50,000	58	0.0171
Mid-range LLM	0.15	0.321	150,000	39	0.0247
Open-source LLM	0.25	0.298	250,000	34	0.0292
Smaller model	0.35	0.276	350,000	30	0.0327

6 The Verification Horizon

Theorem 32 (Verification Horizon)**.** *Let $\varepsilon(N) = c_0 N^{-\alpha}$ be the error rate scaling law and M_{total} the total available labeled data. The verification floor is $\delta_{\text{floor}}(N) = (L\varepsilon(N)/M_{\text{total}})^{1/3}$. Verification is meaningful ($\delta_{\text{floor}}(N) < \varepsilon(N)$) iff $N < N^*$, where*

$$N^* = \left(\frac{c_0^2 M_{\text{total}}}{L} \right)^{1/(2\alpha)}. \tag{24}$$

For $N > N^*$, no verifier can verify calibration to a precision finer than the model's ECE. Active verification (Theorem 24) shifts the horizon to $N^*_{\text{active}} = (c_0 M_{\text{total}})^{1/\alpha}$, which is strictly larger and L-free.

Proof sketch. From Theorem 9, $m \geq L\varepsilon/\delta^3$. With $m = M_{\text{total}}$, the best δ is $\delta_{\text{floor}} = (L\varepsilon/M_{\text{total}})^{1/3}$. Meaningful verification requires $\varepsilon > \sqrt{L/M_{\text{total}}}$; substituting the scaling law gives $N < N^*$. □ □

Quantitative estimate. Under Chinchilla scaling ($\alpha \approx 0.5$, $c_0 \approx 1$) with $L = 1$ and $M_{\text{total}} = 10^7$: $N^* \approx 10^7$ parameters — well below current frontier models ($\sim 10^{11}$). N^* should be interpreted as an order-of-magnitude guide; both α and M_{total} are empirical and subject to uncertainty.

Dynamic and temporal extensions. The verification tax has three dynamic consequences. (i) *Verification dynamics*: if a model's error rate improves as $\varepsilon(t) = c_0 t^{-\beta}$ under deployment-rate r, the critical exponent for verification keeping pace is $\beta^* = 1/2$ (passive) and $\beta^* = 1$ (active). Models improving faster than β^* outrun verification. (ii) *Recalibration trap*: iterative recalibration that halves ECE per round hits a ceiling of $K^* = O(\log M_{\text{total}})$ verifiable rounds. On MMLU ($M_{\text{total}} \approx 14{,}000$, $\varepsilon = 0.05$, $\text{ECE}_0 = 0.10$, $\gamma = 0.5$): only $K^* = 4$ rounds are verifiable. (iii) *Verification half-life*: under ECE drift rate λ, every verification expires after $t_{1/2} = \delta/\lambda$, and perpetual verification requires data rate $r \geq L\varepsilon\lambda/\delta^4$. Domain estimates range from hours (financial) to months (healthcare). Full theorems, proofs, and tables: Appendix C.

7 Compositional Verification

Modern AI systems compose multiple components — retrieval-augmented generation (Lewis et al., 2020), chain-of-thought reasoning (Wei et al., 2022), and agentic loops (Yao et al., 2023). Even if every component is individually well-calibrated, the composed system need not be.

Theorem 33 (Compositional Verification Tax)**.** *For a K-component pipeline with per-component Lipschitz constants $L_1, \ldots, L_K$ (each g_k being L_k-Lipschitz in its confidence input), the end-to-end calibration function has $L_{\text{sys}} \leq \prod_k L_k + 1$ (Lipschitz chain rule; the "+1" arises because $\Delta_{\text{sys}}(p) = \eta_{\text{sys}}(p) - p$ and $p \mapsto p$ is 1-Lipschitz, so $L_\Delta \leq L_\eta + 1 = \prod_k L_k + 1$). Verification then requires $m_{\text{sys}} \geq c \cdot L_{\text{sys}}\varepsilon/\delta^3$. For homogeneous pipelines with $L_k = L > 1$, $m_{\text{sys}} = \Omega(L^K\varepsilon/\delta^3)$ — exponential in pipeline depth. Equivalently, the maximum verifiable depth is $K_{\max} = \lfloor \log_L(M_{\text{total}}\delta^3/\varepsilon) \rfloor$. For $L = 2$, an agent executing 10 reasoning iterations has $1024\times$ the verification cost of a single-step model. Full proof, agent-loop bound, and Lipschitz/contractivity caveats: Appendix D.*

8 Related Work

Calibration estimation. Sun et al. (2023) and Futami & Fujisawa (2024) established $O(n^{-1/3})$ upper bounds on ECE estimation bias. We complement these with a matching lower bound and reveal the ε-dependence absent from their general theory. Ciosek et al. (2025) derive finite-sample bounds under bounded variation but do not study ε-dependence. Hu et al. (2024) study calibration testing (rather than estimation) with $\Omega(\varepsilon^{-2.5})$ sample complexity; Blasiok et al. (2023) introduce calibration distance, a different but related metric. Futami & Nitanda (2026) address uniform convergence of smooth CE.

Active and sequential evaluation. Kossen et al. (2021) propose practical active testing algorithms; our Theorem 24 provides the matching minimax-optimal rate. Shekhar & Ramdas (2024) develop e-value-based anytime-valid sequential tests for fairness with stopping times matching our Theorem 26; we provide the matching lower bound, confirming optimality.

Why ECE is harder than smooth surrogates. Smooth calibration surrogates (smCE, LDTC) replace the non-smooth $|\Delta(p)|$ with differentiable approximations, whose L_2 structure admits $O(\varepsilon^{-2})$ testing complexity (Hu et al., 2024). ECE's absolute-value integrand is an L_1 functional, which forces the harder $m^{-1/3}$ nonparametric rate (Theorem 8). This is not a deficiency of ECE but reflects the harder statistical problem of estimating an L_1 norm versus an L_2 norm; the metric choice determines the verification cost.

Empirical calibration literature. Guo et al. (2017) popularized ECE and temperature scaling; our Proposition 29 provides the theoretical $m^{-1/2}$ rate for their method. Kumar et al. (2019) propose kernel-based calibration measures (whether they attain the minimax rate is open). Nixon et al. (2019) compared binning strategies empirically; Theorem 9 gives the optimal bin count they sought. Vaicenavicius et al. (2019), Widmann et al. (2019), and Davis et al. (2020) respectively motivate calibration testing, multiclass extensions, and clinical drift detection — all of which our framework formalizes via the corresponding theorems.

Nonparametric functional estimation. Our lower bound builds on Lepski et al. (1999) (minimax L_r norm rates in white noise), with the novel observation that the noise variance equals the error rate ε. The Brown–Low equivalence (Brown & Low, 1996) underpins the reduction.

9 Conclusion

The verification tax states that estimating the calibration error of a model with error rate ε requires $\Theta((L\varepsilon/m)^{1/3})$ labeled samples — a cost that grows as models improve, under

meaningful verification ($\delta = o(\varepsilon)$). The tax is universal across calibration, fairness, and robustness (Theorem 16); cannot be circumvented by model access (Theorem 20); cannot be gamed by adversarial miscalibration (Proposition 22); and cannot be avoided by self-verification (Theorem 27). Two avenues reduce the tax: parametric assumptions yield the parametric $1/\sqrt{m}$ rate (Proposition 29); adaptive evaluation eliminates L entirely, achieving $\Theta(\sqrt{\varepsilon/m})$ (Theorem 24). At scale, the tax creates a verification horizon (Theorem 32), a recalibration trap, a half-life, and compounds exponentially in compositional pipelines (Theorem 33).

Limitations. Our results rely on Lipschitz calibration functions; piecewise-Lipschitz extensions and bounded-variation generalizations are open. The minimax lower bound inherits a logarithmic gap from L_1 functional estimation (Lepski et al., 1999). The compositional bound uses worst-case Lipschitz accumulation; contractive components reduce the effective L_{sys} in practice. Sharp constants are known only for the Le Cam bound. Our framework verifies group-conditional calibration; marginal fairness notions are distinct functionals. Our analysis addresses top-label ECE; extending to vector-valued calibration gaps in multiclass settings, and connecting to multicalibration auditing lower bounds where combinatorial group families may increase statistical complexity, is an important open direction. Three open problems: closing the log gap, handling non-Lipschitz pipeline components, and extending to multiclass calibration.

Broader implications. Self-reported calibration claims carry no information-theoretic weight: independent auditing with fresh labeled data is necessary. Regulators should invest in adaptive evaluation protocols and adopt continuous auditing frameworks with re-audit schedules determined by the verification half-life.

Acknowledgments

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A Full Proof of Theorem 7 (Le Cam Lower Bound)

Construct two hypotheses. Fix a classifier whose score distribution places all mass at $p_0 = 1 - \varepsilon$.

- P_0: $Y \mid p(X) = p_0 \sim \mathrm{Bern}(1-\varepsilon)$. Calibrated: $\eta(p_0) = p_0$, so $\mathrm{ECE} = 0$.
- P_1: $Y \mid p(X) = p_0 \sim \mathrm{Bern}(1-\varepsilon-\delta)$. Miscalibrated: $\mathrm{ECE} = \delta$.

Both hypotheses use a constant calibration gap (Lipschitz constant 0), so $\Delta \in \mathcal{F}(L)$ for all $L > 0$. The KL divergence of m i.i.d. samples is:

$$\mathrm{KL}(P_0^m \| P_1^m) = m \cdot \mathrm{KL}(\mathrm{Bern}(1-\varepsilon) \| \mathrm{Bern}(1-\varepsilon-\delta)) = \frac{m\delta^2}{\varepsilon(1-\varepsilon)} \le \frac{m\delta^2}{\varepsilon}$$

where the last inequality uses $1 - \varepsilon \le 1$ and the KL approximation $\mathrm{KL}(\mathrm{Bern}(p) \| \mathrm{Bern}(p+\delta)) \approx \delta^2/(p(1-p))$ (valid for $\delta \le \varepsilon/2$). By Le Cam's lemma (Tsybakov, 2009):

$$R^* \ge \frac{\delta}{2}\left(1 - \mathrm{TV}(P_0^m, P_1^m)\right) \ge \frac{\delta}{2} e^{-\mathrm{KL}(P_0^m \| P_1^m)}$$

where the second inequality uses Bretagnolle–Huber. Setting $\mathrm{KL} = 1$ (i.e., $\delta = \sqrt{\varepsilon/m}$): $R^* \ge (1/2e)\sqrt{\varepsilon/m} \approx 0.184\sqrt{\varepsilon/m}$. □

A.1 Direct Bernoulli Lower Bound (Alternative to Brown–Low)

For readers who prefer a self-contained argument avoiding asymptotic equivalence, we sketch a direct Assouad-type lower bound in the Bernoulli model.

Partition $[0,1]$ into B equal-width bins. Consider the class of calibration functions $\Delta_\omega(p) = h \cdot \omega_b$ for $p \in I_b$, where $\omega = (\omega_1, \ldots, \omega_B) \in \{-1, +1\}^B$ and $h > 0$ is chosen so that Δ_ω is L-Lipschitz (requiring $h \le L/(2B)$). Each Δ_ω has $\mathrm{ECE} = h$ (since $\sum_b (1/B)|h \cdot \omega_b| = h$). For any two configurations ω, ω' differing in one coordinate b, the KL divergence of the m/B observations in bin b is:

$$\mathrm{KL} = \frac{m}{B} \cdot \frac{(2h)^2}{\varepsilon(1-\varepsilon)} \le \frac{4mh^2}{B\varepsilon}$$

By Assouad's lemma (Tsybakov, 2009), the minimax risk satisfies:

$$R^* \ge \frac{Bh}{2}\left(1 - \sqrt{\frac{2mh^2}{B\varepsilon}}\right)$$

Setting $h = c\sqrt{B\varepsilon/m}$ for a small constant c and $B = \lfloor (Lm/\varepsilon)^{1/3} \rfloor$ gives:

$$R^* \ge c' \left(\frac{L\varepsilon}{m}\right)^{1/3}$$

This recovers the $(L\varepsilon/m)^{1/3}$ rate directly from the Bernoulli model, confirming the Brown–Low reduction. The constant c' is weaker than the one obtained via Lepski et al. (1999), but the argument is fully self-contained. □

B Full Proof of Theorem 9 (Matched Upper Bound)

Bias-variance decomposition. With B equal-width bins and m total samples:

1. Bias. Within each bin of width $1/B$, the calibration gap Δ varies by at most L/B (Lipschitz). By the Bridge Lemma, $|\mathrm{ECE}_B - \mathrm{CE}_{\mathrm{smooth}}| \le L/B$.

2. Variance. Each bin b has $n_b \approx m/B$ samples. (Under bounded density $c_\mu \le d\mu/dp \le C_\mu$; equal-mass binning achieves $n_b = m/B$ exactly when μ concentrates.) The empirical calibration gap has variance $\mathrm{Var}(\hat{\Delta}_b) \le \sigma^2/n_b \approx \sigma^2 B/m$, where $\sigma^2 = \max_b \mathbb{E}[\eta(p)(1-\eta(p)) \mid p \in I_b]$. Estimation error: $\mathbb{E}[|\widehat{\mathrm{ECE}}_B - \mathrm{ECE}_B|] \le \sqrt{\sigma^2 B/m}$.

3. Total error. $L/B + \sqrt{\sigma^2 B/m}$.

4. Optimization. Setting $L/B = \sqrt{\sigma^2 B/m}$: $B^* = (L^2 m/\sigma^2)^{1/3}$, total error $(L\sigma^2/m)^{1/3}$. In the rare-error regime $\sigma^2 \approx \varepsilon$, recovering the headline rate. □

C Verification Dynamics, Recalibration Trap, and Half-Life

C.1 Verification Dynamics

Suppose a model is deployed at rate r samples per unit time and $\varepsilon(t) = c_0 t^{-\beta}$. The verification floor at time t is $\delta_{\text{floor}}(t) = (Lc_0/r)^{1/3} t^{-(\beta+1)/3}$. Meaningful verification ($\delta_{\text{floor}}(t) < \varepsilon(t)$) reduces to $(L/(rc_0^2))^{1/3} < t^{(1-2\beta)/3}$. This is satisfied for large t when $\beta < 1/2$, violated when $\beta > 1/2$, and equivalent to $L/(rc_0^2) < 1$ when $\beta = 1/2$. Critical exponent: $\beta^* = 1/2$ (passive), $\beta^*_{\text{active}} = 1$ (active). □

C.2 Recalibration Trap

With $\text{ECE}_K = \gamma^K \text{ECE}_0$, the improvement from round K to $K+1$ is $\gamma^K \text{ECE}_0(1-\gamma)$. Distinguishing requires (Theorem 7) $m_K \geq \varepsilon/(\gamma^{2K}\text{ECE}_0^2(1-\gamma)^2)$. Setting $m_K = M_{\text{total}}$ gives $K^* = \lfloor \log(M_{\text{total}}(1-\gamma)^2\text{ECE}_0^2/\varepsilon)/(2\log(1/\gamma)) \rfloor$.

Worked example (MMLU). $\gamma = 0.5$, $\varepsilon = 0.05$, $\text{ECE}_0 = 0.10$, $M_{\text{total}} = 14{,}000$: $K^* = \lfloor 6.55/1.386 \rfloor = 4$. After 4 halving rounds, MMLU cannot verify further improvement.

C.3 Verification Half-Life

Let ECE drift rate λ satisfy $|\text{ECE}(t) - \text{ECE}(0)| \leq \lambda t$. (i) The verification becomes invalid after $t_{1/2} = \delta/\lambda$. (ii) Perpetual verification requires data rate $r \geq L\varepsilon\lambda/\delta^4$.

Domain estimates (assuming $L = 1$, $\varepsilon = 0.05$): Medical AI ($\lambda = 0.01$/month, $\delta = 0.02$): $t_{1/2} = 2$ months, $r \approx 3{,}100$/month. Content moderation: 6 months, 620/month. Financial trading ($\lambda = 0.1$/week, $\delta = 0.01$): $t_{1/2} = 17$ hours. Autonomous driving: 15 days. Static benchmarks have the same half-life: a benchmark constructed at $t = 0$ decays at rate λ, formalizing the case for periodic refresh.

D Full Proof of Theorem 33 (Compositional Verification)

Lipschitz composition. By the chain rule, if g_k is L_k-Lipschitz in its first argument, then $g_K \circ \cdots \circ g_1$ is $(\prod_{k=1}^K L_k)$-Lipschitz. Since $\Delta_{\text{sys}}(p) = \eta_{\text{sys}}(p) - p$ and $p \mapsto p$ is 1-Lipschitz, $L_{\text{sys}} \leq \prod_k L_k + 1$.

Verification cost. Applying Theorem 9 with $L = L_{\text{sys}}$: $m_{\text{sys}} = \Theta(L_{\text{sys}}\varepsilon/\delta^3)$. For homogeneous $L_k = L$: $L_{\text{sys}} \approx L^K$, giving exponential scaling. □

Agent loop bound. For an agent with K reasoning iterations and $L = 2$: $L^{10} = 1024$, $L^{20} > 10^6$.

Caveats. The Lipschitz composition model is optimistic in two directions: discontinuous components (e.g., discrete retrieval) yield $L_k = \infty$, making end-to-end verification impossible at *any* sample size; conversely, contractive components ($L_k < 1$) reduce the effective L_{sys}. Empirical estimation of L_{sys} via finite-difference approximation is recommended before applying the exponential bound. Independence of per-component calibration gaps is also assumed; correlated miscalibration can either help or hurt depending on structure.

Worked example. A two-stage retriever–classifier with $L = 2$ each gives $L_{\text{sys}} \leq 5$, requiring 5× the data of a single classifier. With $L = 3$: 10×. A three-stage pipeline at $L = 2$: 9×; at $L = 3$: 28×.

E Full Proof of Theorem 8

Step 1: Reduction to nonparametric regression.

Within a window of width w around $p = 1-\varepsilon$, where the score distribution concentrates, the calibration verification problem reduces to estimating the L_1 norm of a regression function. Specifically:

For $p \in [1-\varepsilon-w/2,\, 1-\varepsilon+w/2]$, define $\Delta(p) = \eta(p) - p$. The labels Y_i given $p(X_i) = p$ are $\mathrm{Bern}(\eta(p)) = \mathrm{Bern}(p + \Delta(p))$, with variance $\eta(p)(1-\eta(p)) \approx \varepsilon$ in the rare-error regime.

The ECE restricted to this window is:

$$\mathrm{ECE}_{\mathrm{window}} = \int_{1-\varepsilon-w/2}^{1-\varepsilon+w/2} |\Delta(p)|\, d\mu(p) \tag{25}$$

This is precisely the L_1 norm of Δ on the window, weighted by μ.

Step 2: Identification with L_1 functional estimation in white noise.

By the asymptotic equivalence of nonparametric regression and Gaussian white noise (Brown & Low, 1996; Nussbaum, 1996), estimating $\int |\Delta(p)|dp$ from m observations with per-observation variance ε is asymptotically equivalent to estimating $\int |f(t)|dt$ in the white noise model:

$$dX(t) = f(t)dt + (m/\varepsilon)^{-1/2} dW(t), \quad t \in [0,1] \tag{26}$$

with effective sample size $n_{\mathrm{eff}} = m/\varepsilon$ and $f \in \Sigma(\beta = 1, L)$ (Lipschitz = Hölder with $\beta = 1$).

Step 3: Application of Lepski, Nemirovski & Spokoiny (1999).

By Theorem 2.2 of Lepski et al. (1999), the minimax rate for estimating the L_1 norm of a β-Hölder function in white noise with sample size n is:

$$R^*_{L_1} \geq c \cdot n^{-\beta/(2\beta+1)} / (\log n)^{c'} \tag{27}$$

For $\beta = 1$ (Lipschitz) and $n = n_{\mathrm{eff}} = m/\varepsilon$:

$$R^*_{L_1} \geq c \cdot (m/\varepsilon)^{-1/3} / (\log(m/\varepsilon))^{c'} = c \cdot \left(\frac{\varepsilon}{m}\right)^{1/3} / (\log m)^{c'} \tag{28}$$

Incorporating the Lipschitz constant L (which scales the function class):

$$R^*(m, \varepsilon, L) \geq c_2 \left(\frac{L\varepsilon}{m}\right)^{1/3} / (\log m)^{c_3} \tag{29}$$

Technical conditions:

- *Score-mass concentration:* The score distribution μ must place mass $\geq \rho$ in a window of width w around $1-\varepsilon$. We state this as an assumption. This holds for well-trained classifiers: overconfident models concentrate predictions near the top.
- *The Brown–Low equivalence* requires m sufficiently large relative to the smoothness. The minimum m is $m \geq C(L)/\varepsilon$ for a constant depending on L.
- *Lipschitz constant:* Since $\Delta(p) = \eta(p) - p$ and $p \mapsto p$ is 1-Lipschitz, we have $L_\Delta \leq L_\eta + 1$. We use L throughout for L_Δ.

Clarification on the asymptotic regime. The Brown–Low equivalence is applied in the limit $m \to \infty$ with ε *fixed*. The error rate ε is a property of the model under audit, not an asymptotic parameter. For any fixed $\varepsilon \in (0, 1/2)$, the Bernoulli label variance $\varepsilon(1-\varepsilon) \geq \varepsilon/2 > 0$ is bounded away from zero, satisfying the regularity conditions. The ε-dependence in the rate $(L\varepsilon/m)^{1/3}$ arises from substituting $\sigma^2 = \varepsilon(1-\varepsilon) \approx \varepsilon$ into the white-noise minimax rate with $n_{\mathrm{eff}} = m/\varepsilon$, not from taking $\varepsilon \to 0$ within the equivalence framework.

F Supplementary Details for Theorem 20

The proof of Theorem 20 in the main text is complete. We note that the result extends straightforwardly to any side information $\mathcal{S}$ that is independent of the fresh evaluation labels conditional on the scores. This includes: model weights, architecture specifications, training data, training logs, gradient histories, and any derived quantity (e.g., Fisher information matrices, Hessian spectra, pruning masks). The key insight is that calibration is a property of the model-world interface, not of the model alone.

G Full Proof of Theorem 24 (Active Verification)

Lower bound. The Le Cam lower bound from Theorem 7 applies unchanged to adaptive strategies. Construct the same two hypotheses: P_0 (calibrated, ECE $= 0$) and P_1 (miscalibrated by δ near $p = 1-\varepsilon$, ECE $= \delta$). Under adaptive querying, the auditor selects p_i based on all previous observations $(p_1, y_1, \ldots, p_{i-1}, y_{i-1})$. By the chain rule of KL divergence:

$$\mathrm{KL}(P_0^m \| P_1^m) = \sum_{i=1}^{m} \mathbb{E}\left[\mathrm{KL}(P_0(\cdot \mid p_i) \| P_1(\cdot \mid p_i))\right] \tag{30}$$

Each term is at most δ^2/ε (the per-observation KL when p_i is in the miscalibrated region, zero otherwise). Adaptivity does not increase the per-observation information: conditioned on p_i, the label y_i is a single Bernoulli draw whose KL is δ^2/ε regardless of previous observations. Therefore KL $\leq m\delta^2/\varepsilon$, and Le Cam gives $R^*_{\text{active}} \geq c\sqrt{\varepsilon/m}$.

Upper bound. We construct a two-phase adaptive strategy.

Phase 1 (Exploration): Select N equally-spaced confidence levels $\{p_j = j/N\}_{j=1}^{N}$ in the interval $[1-\varepsilon-w, 1-\varepsilon+w]$ for a window width $w = O(1)$. Query each level $m/(2N)$ times. The empirical estimate $\hat{\Delta}(p_j)$ has standard error $\sigma_j = \sqrt{2\varepsilon N/m}$. Classify each bin as:

- *Resolved*: $|\hat{\Delta}(p_j)| > 2\sigma_j$ — the sign of $\Delta(p_j)$ is known with high probability.
- *Unresolved*: $|\hat{\Delta}(p_j)| \leq 2\sigma_j$ — the sign is uncertain.

The unresolved zone has width at most $4\sigma_j/L = O(\sqrt{\varepsilon N/m}/L)$ by the Lipschitz constraint (the function can only stay near zero for a region of width $\leq |\Delta|_{\max}/L$).

Phase 2 (Exploitation): Use the remaining $m/2$ queries on resolved bins to estimate $|\Delta(p_j)| = \Delta(p_j) \cdot \mathrm{sign}(\Delta(p_j))$ (sign is known, so no absolute value problem). This is a simple average with error $\sqrt{\varepsilon/m}$.

Total error:

- Resolved bins: estimation error $O(\sqrt{\varepsilon/m})$.
- Unresolved bins: the contribution to ECE is at most (max $|\Delta|$ in zone) $\times$ (width of zone). Since $|\Delta| \leq 2\sigma_j$ in the zone and the width is $O(\sigma_j/L)$, this contributes $O(\sigma_j^2/L) = O(\varepsilon N/(mL))$.

Choose $N = L\sqrt{m/\varepsilon}$ to balance the two terms. Both become $O(\sqrt{\varepsilon/m})$. □

H Full Proof of Theorem 26 (Sequential Verification)

Lower bound. Under the two Le Cam hypotheses (P_0: ECE $= 0$, P_1: ECE $= \delta$), the log-likelihood ratio after t observations is:

$$\Lambda_t = \sum_{i=1}^{t} \log \frac{p_1(y_i \mid p_i)}{p_0(y_i \mid p_i)} \tag{31}$$

This is a random walk with per-step drift:

$$\mathbb{E}_{P_0}[\Lambda_1] = -\mathrm{KL}(P_0(\cdot \mid p_i) \| P_1(\cdot \mid p_i)) \approx -\frac{\delta^2}{\varepsilon} \tag{32}$$

By Wald's identity for sequential testing (Tsybakov, 2009), any test between P_0 and P_1 with error probabilities $\leq \alpha$ satisfies:

$$\mathbb{E}[\tau] \geq \frac{\log(1/\alpha)}{\mathrm{KL}(P_0 \| P_1)} = \frac{\varepsilon \log(1/\alpha)}{\delta^2} \tag{33}$$

Since any sequential verification protocol with accuracy δ at confidence $1-\alpha$ must distinguish $\mathrm{ECE} = 0$ from $\mathrm{ECE} = \delta$ with probability $\geq 1 - \alpha$, the lower bound follows.

Upper bound. We construct a confidence sequence for the ECE using the method of mixtures (Howard et al., 2021). Define the running ECE estimate:

$$\widehat{\mathrm{ECE}}_t = \sum_{b=1}^{B} \frac{n_{b,t}}{t} \left| \hat{\Delta}_{b,t} \right| \tag{34}$$

where $\hat{\Delta}_{b,t}$ is the empirical calibration gap in bin b after t observations.

By Theorem 1 of Howard et al. (2021), there exists a time-uniform confidence sequence $(C_t)_{t \geq 1}$ such that:

$$\Pr\left(\exists t \geq 1 : |\widehat{\mathrm{ECE}}_t - \mathrm{ECE}| > C_t\right) \leq \alpha \tag{35}$$

with $C_t = O(\sqrt{\varepsilon \log(\log(t)/\alpha)/t})$.

Define the stopping time $\tau = \inf\{t : C_t \leq \delta\}$. Solving $C_t \leq \delta$:

$$\sqrt{\frac{\varepsilon \log(\log(t)/\alpha)}{t}} \leq \delta \implies t \geq \frac{\varepsilon \log(\log(t)/\alpha)}{\delta^2} \tag{36}$$

For the dominant term, $\mathbb{E}[\tau] = O(\varepsilon \log(1/\alpha)/\delta^2)$, matching the lower bound up to constants. □

I Temperature Scaling Analysis

We provide the full proof of Proposition 29.

Step 1: MLE for the temperature parameter. Given scores $p_1, \ldots, p_m$ and labels $y_1, \ldots, y_m$, the negative log-likelihood is:

$$\ell(T) = -\sum_{i=1}^{m} [y_i \log \sigma(\mathrm{logit}(p_i)/T) + (1 - y_i) \log(1 - \sigma(\mathrm{logit}(p_i)/T))] \tag{37}$$

This is a 1D convex optimization. The MLE $\hat{T}$ is unique.

Step 2: Asymptotic normality of MLE. By standard M-estimation theory (van der Vaart, 1998):

$$\sqrt{m}(\hat{T} - T^*) \xrightarrow{d} N(0, \mathcal{I}(T^*)^{-1}) \tag{38}$$

where $\mathcal{I}(T^*)$ is the Fisher information. At $T^* = 1$, the score function is:

$$\frac{\partial}{\partial T} \log p(Y \mid p, T) \bigg|_{T=1} = -(Y - p) \cdot \mathrm{logit}(p) \tag{39}$$

So:

$$\mathcal{I}(1) = \mathbb{E}[(Y - p)^2 \cdot \mathrm{logit}(p)^2] = \mathbb{E}[p(1 - p) \cdot \mathrm{logit}(p)^2] \tag{40}$$

In the rare-error regime with scores concentrated near $1 - \varepsilon$: $\mathrm{logit}(1 - \varepsilon) \approx \log(1/\varepsilon)$ and $p(1 - p) \approx \varepsilon$, giving:

$$\mathcal{I}(1) \approx \varepsilon \cdot \log^2(1/\varepsilon) \tag{41}$$

This is positive and grows as $\varepsilon \to 0$ — rare errors make temperature estimation *easier* because logits are more spread out. The MLE satisfies: $|\hat{T} - T^*| = O_p(1/\sqrt{m \cdot \mathcal{I}(T^*)})$.

Step 3: Translating to calibration error. By Taylor expansion of $\Delta(p; T)$ around $T^* = 1$:

$$\Delta(p; \hat{T}) \approx \left.\frac{\partial \Delta}{\partial T}\right|_{T=1} \cdot (\hat{T} - 1) = p(1-p) \cdot \text{logit}(p) \cdot (\hat{T} - 1) \tag{42}$$

So the smooth CE is:

$$\text{CE}_{\text{smooth}} = \int |\Delta(p; \hat{T})| \, d\mu(p) \approx |\hat{T} - 1| \cdot \underbrace{\int p(1-p)|\text{logit}(p)| \, d\mu(p)}_{\kappa} \tag{43}$$

where κ is a constant depending on the score distribution. Substituting $|\hat{T} - 1| = O_p(1/\sqrt{m \cdot \mathcal{I}(T^*)})$:

$$\text{CE}_{\text{smooth}} = O_p\left(\frac{\kappa}{\sqrt{m \cdot \mathcal{I}(T^*)}}\right) = O_p\left(\frac{1}{\sqrt{m}}\right) \tag{44}$$

with a constant that depends on ε through $\mathcal{I}(T^*)$ and κ. This is the **parametric rate** $m^{-1/2}$, not the nonparametric rate $(L\varepsilon/m)^{1/3}$.

Step 4: Connection to binned ECE. Via the Bridge Lemma: $\text{ECE}_B \leq \text{CE}_{\text{smooth}} + L/B$. Taking $B = O(\sqrt{m}/L)$:

$$\text{ECE}_B = O(1/\sqrt{m}) \tag{45}$$

Comparison to nonparametric rate: The ratio is $(1/\sqrt{m})/(L\varepsilon/m)^{1/3} = (L\varepsilon)^{-1/3}/m^{1/6}$, which shrinks with m. For $m = 10{,}000$ and $L\varepsilon = 0.05$: ratio ≈ 0.05. Temperature scaling is $\sim$20$\times$ more data-efficient than nonparametric ECE estimation. □

J Verification Transfer Details

We provide the full proof of Proposition 31.

Step 1: Problem reduction. Since $\Delta_1(p)$ is known (from prior verification), estimating $\text{ECE}(M_2) = \int |\Delta_2(p)| \, d\mu(p)$ reduces to estimating the shift $h(p) = \Delta_2(p) - \Delta_1(p)$, then computing $\int |\Delta_1(p) + \hat{h}(p)| \, d\mu(p)$.

Step 2: Lower bound (detection cost). Two-sample Le Cam between H_0: $\Delta_2 = \Delta_1$ and H_1: $\Delta_2 = \Delta_1 + h$ with $\|h\|_\infty = \delta_{\text{ft}}$. From m_2 fresh samples near $p \approx 1 - \varepsilon_2$:

$$\text{KL}(H_0^{m_2} \| H_1^{m_2}) \approx \frac{m_2 \delta_{\text{ft}}^2}{\varepsilon_2} \tag{46}$$

Indistinguishable when KL ≤ 1, requiring $m_2 = \Omega(\varepsilon_2/\delta_{\text{ft}}^2)$.

Step 3: Upper bound (estimation with transfer). Apply Theorem 9 to h instead of Δ_2. The histogram estimator for h has bias L_h/B and variance $\sqrt{\varepsilon_2 B/m_2}$. Optimizing: $B^* = (L_h^2 m_2/\varepsilon_2)^{1/3}$, giving:

$$\|\hat{h} - h\| = O\left(\left(\frac{L_h \varepsilon_2}{m_2}\right)^{1/3}\right) \tag{47}$$

By the triangle inequality:

$$\left|\text{ECE}(M_2) - \widehat{\text{ECE}}(M_2)\right| \leq \int |\hat{h}(p) - h(p)| \, d\mu(p) = O\left(\left(\frac{L_h \varepsilon_2}{m_2}\right)^{1/3}\right) \tag{48}$$

Setting this equal to δ_{ft}: $m_2 = O(L_h \varepsilon_2/\delta_{\text{ft}}^3)$.

Step 4: When does transfer help? From-scratch: $m_2 = \Theta(L\varepsilon_2/\delta^3)$. With transfer: $m_2 = \Theta(L_h \varepsilon_2/\delta_{\text{ft}}^3)$. Transfer helps when $\delta_{\text{ft}} < \delta$ and $L_h \leq L$. With $\delta_{\text{ft}} = \delta/10$, saving is 1000$\times$. Transfer provides no benefit when $\delta_{\text{ft}} \geq \delta$. □

K Corrections: Correlation, Heterogeneity, Pre-Calibration

The core theorems assume i.i.d. labeled samples with independent Bernoulli noise. In practice, several deviations may arise.

Temporal correlation. If evaluation samples arrive sequentially and exhibit temporal dependence, the effective sample size is reduced by a factor depending on the mixing time. For β-mixing sequences with coefficient $\beta(k) \leq c \cdot \rho^k$, the effective sample size is $m_{\text{eff}} \approx m \cdot (1 - \rho)$. All bounds apply with m replaced by m_{eff}.

Domain heterogeneity. When the evaluation set spans multiple domains with different error rates, the per-domain verification floor (Fix 3) accounts for this heterogeneity. The overall ECE is a weighted average of domain-specific ECEs, and Jensen's inequality ensures the overall verification floor is no larger than the worst per-domain floor weighted by domain proportion.

Pre-calibration via RLHF. Models that have undergone reinforcement learning from human feedback may have artificially smooth calibration functions (small L), which reduces the verification tax. However, if the RLHF procedure oversmooths, the remaining miscalibration may be concentrated at specific confidence levels, violating the uniform Lipschitz assumption. Domain-specific Lipschitz estimates should be used in such cases.

L Numerical Sanity Checks

L.1 Sharp Le Cam Constants

For each ε, we numerically optimize the Le Cam lower bound over the separation parameter δ, using the exact Bernoulli KL divergence (not the quadratic approximation). We compute three versions: the Bretagnolle–Huber bound $R^* \geq (\delta/2)\exp(-\text{KL})$, the Pinsker bound $R^* \geq (\delta/2)(1-\sqrt{\text{KL}/2})$, and the exact bound via the Bernoulli likelihood ratio test. Table 3 reports the tightest constant for each ε.

Table 3: Sharp Le Cam constants $c_1(\varepsilon)$ and nonparametric estimation sample sizes. The Le Cam constant governs detection; the estimation column uses $m \geq L\varepsilon/\delta^3$ with $L = 1$, $\delta = 0.01$.

ε	$c_1(\varepsilon)$	m_{detect} (Le Cam)	m_{estimate} (nonparametric)
0.01	0.3780	15	10,000
0.02	0.3638	27	20,000
0.05	0.3474	61	50,000
0.10	0.3326	111	100,000
0.15	0.3206	155	150,000
0.20	0.3093	192	200,000
0.25	0.2982	223	250,000
0.30	0.2873	248	300,000
0.35	0.2759	267	350,000

The constant $c_1(\varepsilon)$ is monotonically decreasing in ε: smaller error rates yield tighter constants because the Bernoulli variance is smaller and the KL approximation is sharper. All values fall in $[0.28, 0.38]$, consistent with the theoretical prediction $c_1 \geq 1/(2e) \approx 0.184$ from the quadratic KL approximation. The exact constants are roughly $2\times$ larger, reflecting the tightness gained from exact Bernoulli KL computation.